Right-Turn Spectrum Languages: One-Pebble Completeness and Deterministic Dilation Obstructions

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Abstract

Consider finite ordered full binary trees with exactly three marked leaves, and let $\kappa(t)$ count the right-child edges between the common ancestor of all three marked leaves and the common ancestor of the last two. For a numerical spectrum $E \subseteq \mathbb{N}$, put $\mathcal{L}_E = \{t : \kappa(t) \in E\}$. We establish three classifications within this family. First, $\mathcal{L}_E$ is a regular tree language precisely when E is ultimately periodic; we give an explicit bottom-up recognizer and matching linear-order upper and lower state-complexity bounds in its preperiod and eventual period. Second, using the three-marker separation of Bojańczyk and Colcombet and an inverse finite-control simulation of right-edge dilation, we derive a criterion excluding $\mathcal{L}_E$ from deterministic tree-walking recognition. It implies the nonrecognizability of every fixed upper or lower threshold, as well as several infinite ultimately periodic spectra. For ultimately periodic E, we identify exactly when arbitrary uniform dilation preimages and finite Boolean combinations recover the original three-marker separator. Third, every regular member of this family has a deterministic one-pebble tree-walking recognizer, providing an explicit zero-/one-pebble contrast. The unannotated, zero-pebble deterministic recognizability of the pure even spectrum is not decided by these results and is stated as a focused question. The research originates in a semantic-versus-physical realization program developed in an earlier monograph, but all automata-theoretic proofs below stand independently of that framework. We distinguish established proof inputs from the parameterized statements established here.

Keywords: tree-walking automata; deterministic nonrecognizability; pebble automata; regular tree languages; right-edge dilation; unary periodicity.

1 Introduction and contribution boundaries

A tree-walking automaton explores a finite input tree using one movable head and finite-state control. Bojańczyk and Colcombet proved that deterministic tree-walking automata are strictly weaker than nondeterministic ones, with a three-marked-leaf witness [2]. Subsequent results showed that even nondeterministic tree-walking automata fail to capture all regular tree languages [4], and nondeterministic recognition is not closed under complementation [9]. The deterministic model is closed under complementation [5].

We take the original three-leaf separation as a starting point, rather than treating it as a new theorem. By measuring the number of right edges on one distinguished path between branching points, we obtain a parameterized family of languages indexed by $E \subseteq \mathbb{N}$. We ask three different questions: which spectra make the language regular, which regular spectra can be eliminated by a uniform dilation reduction from the classical witness, and which members become recognizable with one visible pebble.

The resulting statements are *family-specific*. They are not a general algorithm deciding DTWA membership; even separation by a deterministic tree-walking automaton is undecidable in general [8]. Our use of inverse simulation is also related to known inverse-homomorphism closure results for graph-walking automata [7]. The emphasis of this manuscript is an explicit, jointly organized family of consequences, not a claimed new foundational automata separation. In particular, the existence of languages distinguishable by a deterministic one-pebble walker but not an ordinary walker is not itself new: the older pebble literature already contains such examples [6, 3]. The scope of the remaining pure-parity question is made precise in Section 6.

Research lineage and mathematical provenance

The present problem arose while testing the semantic/physical distinction developed in the author's monograph *Semantic Normalization and Physical Realization* [1] against the unambiguous-versus-deterministic tree-walking question raised in the recent literature [9]. Chapters 108–109 of the monograph distinguish a unique physical outcome for each semantic input from the existence of a single uniform, transition-closed realization, and then study the assembly of protected local obligations inside one model. This distinction motivated the search for concrete tree-walking examples in which local evidence does not compile into one deterministic traversal.

That provenance is *conceptual, not a hidden proof dependency.* The results of this article are proved by finite-tree constructions, classical tree-walking separation, and explicit one-pebble traversal. In particular, they do not assert a UTWA–DTWA separation, do not use a general USNRT theorem to deduce a tree-walking lower bound, and do not claim that the monograph itself proves the statements here.

Which strict inclusions are known?

The ambient recognition classes satisfy

$$\mathsf{DTWA} \subseteq \mathsf{UTWA} \subsetneq \mathsf{NTWA} \subsetneq \mathsf{REG}, \qquad \mathsf{DTWA} \subsetneq \mathsf{NTWA}. \tag{1}$$

The first containment is definitional. The strict deterministic/nondeterministic separation is due to Bojańczyk–Colcombet [2]; the strict unambiguous/nondeterministic separation is due to Martynova–Okhotin [9]; and ordinary nondeterministic walkers do not recognize every regular tree language [4]. Whether the first containment in (1) is strict is a *separate unresolved question in the cited 2025 work*, not a theorem claimed in this article.

The one-pebble class 1pDTWA changes the machine model and should not be inserted as an intermediate class between UTWA and NTWA. Our bounded-threshold examples give an explicit $\mathsf{DTWA} \subsetneq \mathsf{1pDTWA}$ witness, but do *not* thereby produce a witness in $\mathsf{UTWA} \setminus \mathsf{DTWA}$. Nor do the proofs below establish ordinary NTWA recognizability of every threshold language. The three-marker branching language $\mathcal{B}$ is the established $\mathsf{NTWA} \setminus \mathsf{DTWA}$ witness; other claimed lower bounds are stated only in the classes that their proofs support.

2 Three-marker trees and the regularity boundary

Let $\mathcal{W}_3$ be the set of finite ordered full binary trees whose internal nodes are blank and whose leaves are labelled a or b, with exactly three a-labelled leaves. Denote those three leaves, in left-to-right order, by a_1, a_2, a_3. A basic deterministic tree-walking automaton recognizes $\mathcal{W}_3$: perform a standard complete depth-first tour and saturate the number of marked leaves at four.

For $t \in \mathcal{W}_3$, define

$$r(t) = \mathrm{lca}(a_1, a_2, a_3), \qquad q(t) = \mathrm{lca}(a_2, a_3), \qquad \kappa(t) = \#\{\text{right-child edges on } r(t) \rightsquigarrow q(t)\}. \tag{2}$$

The downward path can be empty. In the left-pair branching case $((a_1, a_2), a_3)$, $q = r$ and $\kappa = 0$; in the right-pair case $(a_1, (a_2, a_3))$, q lies in the right subtree of r, so $\kappa \geq 1$. This remains true with arbitrary blank subtrees inserted. For $E \subseteq \mathbb{N} = \{0, 1, 2, \ldots\}$, define

$$\mathcal{L}_E = \{t \in \mathcal{W}_3 : \kappa(t) \in E\}, \quad \mathcal{Z} = \mathcal{L}_{\{0\}}, \quad \mathcal{B} = \mathcal{L}_{\{1,2,3,\ldots\}}.$$

The three-marker separation from [2] is $\mathcal{B} \notin \mathsf{DTWA}$. Because $\mathcal{W}_3 \in \mathsf{DTWA}$ and deterministic tree-walking languages are Boolean-closed [5], also $\mathcal{Z} \notin \mathsf{DTWA}$.

A set $E \subseteq \mathbb{N}$ is *ultimately periodic* if there are $N \geq 0$ and $p \geq 1$ with $\mathbf{1}_E(n + p) = \mathbf{1}_E(n)$ for all $n \geq N$.

Theorem 2.1 (Exact regularity characterization)**.** *For every $E \subseteq \mathbb{N}$,*

$$\boxed{\mathcal{L}_E \in \mathsf{REG} \iff E \text{ is ultimately periodic.}}$$

If N, p witness ultimate periodicity, a deterministic bottom-up automaton with at most $3 + 2(N + p)$ states can be constructed.

Proof. For necessity let $D = \mathrm{Node}(a, a)$, $C[z] = \mathrm{Node}(b, z)$, and $R[z] = \mathrm{Node}(a, z)$, where each displayed b is an unmarked leaf. For $n \geq 0$ let $T_n = R[C^n[D]]$. The three marked leaves are ordered as required and $\kappa(T_n) = n + 1$. In any deterministic bottom-up finite automaton, the state reached on $C^n[D]$ evolves by iterating a single self-map of a finite state set. Its state sequence is eventually periodic. Acceptance of the fixed outer context $R[-]$ is then eventually periodic, so $\mathbf{1}_E(n + 1)$, and thus $\mathbf{1}_E(n)$, is ultimately periodic.

Conversely, choose an ultimately periodic representation with threshold N and period p. Let $\rho(v)$ retain v exactly for $v < N$ and retain the residue $v \bmod p$ in a disjoint periodic part for $v \geq N$. The domain $\mathcal{D}_{N,p}$ contains $N + p$ values, supports a deterministic increment, and determines whether $v \in E$.

At each subtree, store its number c of marked leaves capped at four. When $c = 2$, additionally store $\rho(d)$, where d is the number of right-child edges from the subtree root to the LCA of its two marked leaves. When $c = 3$, instead store $\rho(\kappa)$ for its three marked leaves. The other three count classes $c \in \{0, 1, \geq 4\}$ need no auxiliary value. The complete transition rules at an internal node with left and right child counts c_L, c_R are

count split for $c = 2$	$(1,1)$	$(2,0)$	$(0,2)$
d_{new}	0	d_L	$1 + d_R$

count split for $c = 3$	$(3,0)$	$(0,3)$	$(2,1)$	$(1,2)$
κ_{new}	κ_L	κ_R	0	$1 + d_R$

The additions are performed in $\mathcal{D}_{N,p}$. At the root accept precisely if the marked-leaf count is three and the represented statistic belongs to E. There are at most $3 + 2(N + p)$ states. □

Theorem 2.2 (Linear-order bottom-up state complexity)**.** *Suppose E is ultimately periodic. Let $p \geq 1$ be its* least eventual period*, and let $N \geq 0$ be the least threshold from which membership is periodic with period p. If s_E denotes the minimum number of states of a deterministic bottom-up tree automaton recognizing $\mathcal{L}_E$, then*

$$\boxed{p + \max\{N - 1, 0\} \;\leq\; s_E \;\leq\; 3 + 2(N + p).}$$

Consequently the deterministic bottom-up state complexity of this family has tight linear order *in $N+p$; this is not a lower bound on the state count of a deterministic tree-*walking *automaton.*

Proof. The upper bound is the explicit construction in Theorem 2.1. For the lower bound retain the fixed binary tree $D = \mathrm{Node}(a, a)$ and unary context $C[z] = \mathrm{Node}(b, z)$ from that proof. If a deterministic bottom-up recognizer has state set Q, the state u_n at $C^n[D]$ is obtained from u_0 by n iterations of a *single* map $f : Q \to Q$. Its acceptance by the fixed outer context $R[z] = \mathrm{Node}(a, z)$ is precisely

$$\mathbf{1}_E(n+1) \qquad (n \geq 0).$$

The state orbit of a finite map has an initial tail of length μ followed by a cycle of length λ, using exactly $\mu + \lambda$ distinct states. The displayed Boolean output sequence then has eventual period dividing λ and becomes periodic no later than index μ. Its *least* eventual period is p, while its least preperiod is $\max\{N-1, 0\}$: shifting a unary sequence by one index removes at most its initial exceptional value. Indeed, any periodicity of the shifted sequence from an earlier index, possibly with a nonminimal period, would extend the original eventually-p periodic tail back to an index earlier than N. Thus $p \mid \lambda$ and $\mu \geq \max\{N-1, 0\}$. Therefore

$$|Q| \geq \mu + \lambda \geq p + \max\{N-1, 0\}.$$

This holds for every deterministic bottom-up recognizer and hence for the minimum s_E. □

Corollary 2.3 (Growing numerical thresholds need growing bottom-up control)**.** *For the threshold spectrum $E = \{0, \ldots, B\}$ with $B \geq 1$, a deterministic bottom-up recognizer of $\mathcal{L}_E$ needs at least $B + 1$ states and the construction of Theorem 2.1 uses at most $2B + 7$ states. For the pure modulus-m spectrum $E = m\mathbb{N}$, $m \geq 2$, every such recognizer needs at least m states.*

Proof. For the threshold spectrum the least preperiod is $N = B + 1$ and eventual period is $p = 1$; for $m\mathbb{N}$ they are $N = 0$ and $p = m$. Substitute in Theorem 2.2. □

Corollary 2.4. *If E is not ultimately periodic, $\mathcal{L}_E$ is not recognized by any ordinary or finite-pebble tree-walking automaton. In particular it is not in* DTWA*.*

Proof. Ordinary tree-walking languages are regular [2, 4]; languages of the standard nested finite-pebble model are regular [6]. Apply Theorem 2.1. □

3 Right-edge dilation and a nonrecognizability criterion

For $d \geq 1$, define $R_0(z) = z$, $R_{j+1}(z) = \mathrm{Node}(b, R_j(z))$ and the tree transformation

$$H_d(a) = a, \quad H_d(b) = b, \qquad H_d(\mathrm{Node}(u, v)) = \mathrm{Node}\big(H_d(u), R_{d-1}(H_d(v))\big). \tag{3}$$

Every original right-child edge becomes a chain of d right-child edges with blank left sibling leaves; original left-child edges are unchanged.

Lemma 3.1 (Exact dilation)**.** *H_d preserves the number of marked leaves and, for every $t \in \mathcal{W}_3$, $\kappa(H_d(t)) = d\,\kappa(t)$.*

Proof. The marked-leaf order and the corresponding original lowest common ancestors are unchanged. The path joining the two relevant ancestor nodes is obtained from the old path by replacing each right-child edge with exactly d such edges; a left-child edge remains left-child. □

Lemma 3.2 (Finite-control inverse simulation)**.** *For fixed $d \geq 1$, $K \in$ DTWA implies $H_d^{-1}(K) \in$ DTWA.*

Proof. A machine reading t can simulate a deterministic walker on the virtual tree $H_d(t)$ without constructing that tree. Encode a virtual head position by a physical input node plus one of finitely many tags: an original node, one of the $d-1$ wrapper nodes replacing a right edge out of the current original node, or one of the corresponding blank left leaves. For a wrapper node or its blank left leaf, the physical position is the original parent of the expanded edge. Local parent/left/right/stay transitions of the virtual machine translate into a bounded number of physical movements and tag changes, determined by local input node type and the fixed d. Labels and node types in the virtual tree can also be recovered from this representation. Consequently, the product of the original finite control with the tag set supplies a deterministic simulation. This is a direct special-case inverse transformation argument, consonant with [7]. □

Theorem 3.3 (Spectrum dilation obstruction)**.** *Let $E \subseteq \mathbb{N}$. If some $d \geq 1$ satisfies*

$$\mathbf{1}_E(dn) = 1 - \mathbf{1}_E(0) \qquad (n \geq 1), \tag{4}$$

then $\mathcal{L}_E \notin \mathsf{DTWA}$.

Proof. Inside $\mathcal{W}_3$, Lemma 3.1 identifies $H_d^{-1}(\mathcal{L}_E)$ with $\mathcal{Z}$ if $0 \in E$ and with $\mathcal{B}$ otherwise. Either case is outside DTWA by the classical separation and Boolean closure. Were $\mathcal{L}_E$ deterministically recognizable, Lemma 3.2 would recognize this preimage, a contradiction. □

Corollary 3.4 (Uniform threshold nonrecognizability)**.** *For every fixed $B \geq 0$,*

$$\boxed{\mathcal{L}_{\{0,\ldots,B\}}, \quad \mathcal{L}_{\{B+1,B+2,\ldots\}} \in \mathsf{REG} \setminus \mathsf{DTWA}.}$$

Proof. For each spectrum use $d = B+1$ in Theorem 3.3, and apply Theorem 2.1 for regularity. □

Proposition 3.5 (Sharp factor within uniform dilation)**.** *For every $d \geq 1$ and $B \geq 0$,*

$$H_d^{-1}(\mathcal{L}_{\{0,\ldots,B\}}) = \mathcal{L}_{\{0,\ldots,\lfloor B/d \rfloor\}}.$$

Consequently, the least dilation factor whose inverse image is exactly $\mathcal{Z}$ is $B+1$.

Proof. The inequality $d\,\kappa(t) \leq B$ is equivalent to $\kappa(t) \leq \lfloor B/d \rfloor$. This bound is zero precisely when $d > B$. Every nonnegative value of κ occurs in $\mathcal{W}_3$, so smaller d cannot give exactly $\mathcal{Z}$. □

Corollary 3.6 (Periodic spectra, including infinite examples)**.** *For every $m \geq 2$ and $R \subseteq \{0, \ldots, m-1\}$ not containing 0, the spectrum*

$$E = \{0\} \cup \{n \geq 1 : n \bmod m \in R\}$$

satisfies $\mathcal{L}_E \in \mathsf{REG} \setminus \mathsf{DTWA}$. The same holds for $E = \{n \geq 1 : n \bmod m \in R\}$ whenever $0 \in R$.

Proof. Set $d = m$ in Theorem 3.3 and use ultimate periodicity for regularity. In the first displayed family the choice $R = \varnothing$ gives the finite spectrum $\{0\}$; to obtain an infinite spectrum there take $R \neq \varnothing$. □

4 Exactly what the dilation method proves

For ultimately periodic E with threshold N and period p, put

$$b_E = \mathbf{1}_E(0), \qquad e_E = \mathbf{1}_E(np) \quad \text{for sufficiently large } n.$$

Let $\mathcal{A}_E$ be the Boolean algebra of subsets of $\mathcal{W}_3$ generated by $H_d^{-1}(\mathcal{L}_E)$ for all positive integers d, with complements taken relative to $\mathcal{W}_3$. For clarity, this construction allows *only* these inverse images and finite Boolean operations; it does not encompass arbitrary DTWA proof techniques.

Theorem 4.1 (Exact reach of the dilation–Boolean method)**.** *For every ultimately periodic E, the following are equivalent:*

(i) $b_E \neq e_E$*;*
(ii) *for some $d \geq 1$, $H_d^{-1}(\mathcal{L}_E)$ is $\mathcal{Z}$ or $\mathcal{B}$;*
(iii) $\mathcal{Z} \in \mathcal{A}_E$ *(equivalently, $\mathcal{B} \in \mathcal{A}_E$).*

Proof. If $b_E \neq e_E$, choose $d \geq N$ divisible by p. At $\kappa = 0$, the inverse image has membership b_E; at every positive κ, $d\kappa$ is beyond the threshold and congruent to 0 modulo p, so the membership is $e_E = 1 - b_E$. This proves (i)⇒(ii), and (ii)⇒(iii) follows by Boolean closure.

For the converse suppose $b_E = e_E$. Choose a positive multiple q of p with $q \geq N$. For every $d \geq 1$,

$$\mathbf{1}_E(d \cdot 0) = b_E = e_E = \mathbf{1}_E(dq).$$

Thus each generator, and every Boolean combination of them, agrees on the nonempty fibers $\kappa = 0$ and $\kappa = q$. Neither $\mathcal{Z}$ nor $\mathcal{B}$ does. Therefore (iii) fails. □

Remark 4.2 (A method frontier, not a class characterization)**.** *Theorem 4.1 says exactly when this particular construction can recover the known branching separator. If $b_E = e_E$, it does* not *follow that $\mathcal{L}_E$ belongs to* DTWA*. In particular the pure even spectrum survives this test. The saturated-pattern arguments of [2] alone do not supply a parity-changing equivalent pair by the reductions used here.*

5 One visible pebble recognizes every regular spectrum

In the standard deterministic one-pebble tree-walking model, the machine has a single movable head and finite control, and can place one visible pebble at its current node, detect it when revisiting that node, and lift it there. The pebble must be lifted before acceptance. Let 1pDTWA denote the resulting language class. The nested-pebble regularity inclusion and early anchored-walk techniques were established by Engelfriet and Hoogeboom [6]; a later account of deterministic one-pebble branching tests is given in [3]. The following theorem is a classification of our *particular* numerical family, not the first known zero-/one-pebble class gap.

Lemma 5.1 (Bounded subtree probe)**.** *Given its head at an internal node v, a deterministic one-pebble walker can count, up to any fixed bound, the marked leaves in either chosen child subtree and return to v with its pebble removed.*

Proof. Place the pebble at v and enter the chosen child. Execute a complete local depth-first tour with a finite saturated count. The first upward return to the pebble identifies the boundary v; lift the pebble and resume. The local DFS requires no unbounded stack since the parent/left/right node type is visible at each step. □

Theorem 5.2 (Exact deterministic one-pebble characterization)**.** *For every $E \subseteq \mathbb{N}$,*

$$\boxed{\mathcal{L}_E \in \mathsf{1pDTWA} \iff \mathcal{L}_E \in \mathsf{REG} \iff E \text{ is ultimately periodic.}}$$

Proof. Assume E is ultimately periodic. Begin with a deterministic full-tree DFS verifying membership in $\mathcal{W}_3$. On failure reject. On success, start a fresh preorder DFS, with a finite counter of the marked leaves already visited. At each internal node v, probe both child subtrees via Lemma 5.1; the global preorder counter is untouched during the probes.

If the global counter is one and each child subtree contains exactly one marked leaf, then $v = q(t)$: the visited leaf is a_1 and the two unvisited marked leaves are separated at v. Conversely, in the right-pair case this situation occurs at q, and in the left-pair case no node satisfies it. In the latter case $r = q$, hence $\kappa = 0$, and the machine decides according to $0 \in E$.

In the right-pair case, suspend preorder at q and move upwards. On an ascent from a *left* child, do not probe that parent and do not increment the statistic. On an ascent from a *right* child, increment the finite counter and probe the left subtree of the parent v. This distinction is essential: after a left-child ascent below r, the parent's left subtree can contain a_2, a_3, whereas for a right-child ascent strictly below r the left sibling subtree has no marked leaf. At r the path from r to q begins with a right-child edge and the left subtree contains a_1; therefore the first positive probe performed after a right-child ascent identifies r. The increments are exactly the right-child edges on $q \rightsquigarrow r$, read downward, so their number is $\kappa(t)$.

Implement the counter by exact values below the threshold N and residues modulo p after that threshold, as in Theorem 2.1; increments are monotone on the ascent, and the finite-state summary determines membership in E. The pebble is always lifted after each subtree probe, so only one pebble is used. If the acceptance convention requires a final visit to the root, move there while retaining the decision in one state bit.

Conversely, every language recognized by the standard one-pebble model is regular [6], and Theorem 2.1 then forces E to be ultimately periodic. □

Corollary 5.3 (Explicit deterministic pebble gap)**.** *For every fixed $B \geq 0$,*

$$\boxed{\mathcal{L}_{\{0,\dots,B\}} \in \mathsf{1pDTWA} \setminus \mathsf{DTWA}.}$$

Proof. Combine Theorem 5.2 and Corollary 3.4. □

6 The pure even spectrum: resolved and unresolved aspects

The spectrum $E_{\text{even}} = 2\mathbb{N}$ is purely periodic with period two, and contains zero. Hence

$$\mathcal{L}_{\text{even}} = \{t \in \mathcal{W}_3 : \kappa(t) \equiv 0 \pmod 2\} \in \mathsf{REG} \cap \mathsf{1pDTWA} \tag{5}$$

by Theorems 2.1 and 5.2. Its membership is also unproblematic for a different *zero-pebble* model that accepts according to the parity of the number of accepting nondeterministic computations; this is not ordinary deterministic acceptance.

Lemma 6.1 (Right-turn witness bijection)**.** *On every $t \in \mathcal{W}_3$, the number of left-to-right DFS turns $v : v_L \to v_R$ made after exactly one marked leaf has been visited and with a_3 in the right subtree of v is precisely $\kappa(t)$.*

Proof. If $\kappa(t) = 0$, the leftmost two marked leaves lie on the same side of the top branching node and no such right-subtree turn occurs after seeing only a_1. Otherwise the a_2, a_3 pair lies in the right child subtree of r, and its common ancestor is q. A successful turn's right subtree contains a_3 and occurs after exactly a_1 was visited; it must therefore also contain a_2 (which

precedes a_3) and cannot contain a_1. Thus the turning node is an ancestor of q at which the path to q follows the right child, and is at or below r. Conversely, every right-child edge on the path $r \rightsquigarrow q$ produces exactly one such turn: at that time a_1 has been visited, whereas a_2, a_3 remain in the corresponding right subtree. This gives the bijection. □

Proposition 6.2 (Exact ambiguity and an auxiliary parity-of-runs model)**.** *There is an ordinary nondeterministic tree-walking automaton A_{turn} with no pebbles such that*

$$\#\operatorname{Acc}_{A_{\mathrm{turn}}}(t) = \begin{cases} \kappa(t), & t \in \mathcal{W}_3, \\ 0, & t \notin \mathcal{W}_3. \end{cases}$$

Consequently an automaton with odd-number-of-accepting-runs semantics and no pebbles recognizes $\mathcal{L}_{\mathrm{even}}$. This conclusion is not *deterministic recognizability.*

Proof. For completeness, a zero-pebble nondeterministic walker can be arranged to have exactly $\kappa(t)$ accepting runs on $t \in \mathcal{W}_3$. After checking the three-leaf promise, it performs a deterministic left-to-right DFS. At each transition from a left subtree to the corresponding right subtree, provided exactly one marked leaf has been seen, it may guess that turn, then nondeterministically descend to a marked leaf in the right subtree. Once a marked leaf z is reached, the walker continues the ordinary left-to-right depth-first tour from z to the end of the tree. This continuation starts in the *leave-leaf* phase and accepts the guess precisely when no later marked leaf is encountered. Thus z must be the rightmost marked leaf a_3. Both the successor tour and its one-bit flag for a later marked leaf use finite control, require no pebble, and have no further branching. For a fixed candidate turn, the unique downward path to a_3 is therefore the only successful guessed path, if a_3 lies in its right subtree. By Lemma 6.1, the successful turns are in bijection with the right edges on $r \rightsquigarrow q$, giving $\kappa(t)$ accepting runs. Adding one independent accepting run yields $1 + \kappa(t)$ runs, odd precisely when $\kappa(t)$ is even. No pebble is involved, but the acceptance semantics is *parity-of-runs*, not DTWA. □

The exact remaining question within the family is therefore:

Question 6.3 (Unannotated, zero-pebble deterministic parity)**.** *Is $\mathcal{L}_{\mathrm{even}}$ recognized by an ordinary deterministic tree-walking automaton with one head, no pebbles, and standard single-run acceptance?*

$$\boxed{\mathcal{L}_{\mathrm{even}} \overset{?}{\in} \mathsf{DTWA}.}$$

Theorem 4.1 explains why uniform dilation and Boolean combinations do not decide Question 6.3: here $b_E = e_E = 1$. The one-pebble construction does not decide it either, since its reusable subtree probe relies on an explicitly marked return point. The preceding parity-of-runs construction proves membership only in a different computational model. These are precise limitations of the present arguments, *not* an assertion of general literature-wide openness.

7 Discussion, limitations and future work

The family $\{\mathcal{L}_E : E \subseteq \mathbb{N}\}$ illustrates how a simple numerical statistic can separate three questions: regularity, zero-pebble deterministic recognition, and one-pebble deterministic recognition. Exact regularity and one-pebble membership both reduce to ultimate periodicity. The dilation obstruction transfers the classical three-leaf deterministic lower bound to fixed thresholds and many infinite spectra; its exact Boolean closure boundary is expressible by the value of the spectrum at zero and at sufficiently large multiples of an eventual period.

The proof dependencies matter. The fundamental three-leaf nonrecognizability is due to [2]; Boolean closure and inverse-transformation principles are established machinery [5, 7]; finite-pebble regularity and the general utility of pebbles were known by [6, 3]. The explicit 2025 one-pebble example [9, Theorem 20] concerns a different language; it is not a previous proof of our full spectrum classification. This manuscript specializes, combines, and sharpens those tools for an explicit right-turn family. The contribution claimed here is the explicit spectrum classification, its dilation criterion, the associated finite-state bounds, and the family-specific one-pebble construction. The lower bounds transfer a published separation; they do not establish a new baseline separation of automata classes.

This article leaves the separate DTWA–UTWA equality question untouched: a language outside DTWA is not a counterexample to that equality unless it is also shown to belong to UTWA.

Future work is deliberately limited to Question 6.3: either a concrete ordinary zero-pebble DTWA for $\mathcal{L}_{\text{even}}$ or a proof that every such device fails on some parity-distinguishing tree pair. No positive or negative answer to that question is claimed here, and its resolution is not a prerequisite for the theorems proved in this manuscript.

A small comparison of the established boundaries. For the fixed threshold spectrum $E = \{0, \dots, B\}$, $\mathcal{L}_E$ is regular and one-pebble recognizable but not in DTWA. For the pure even spectrum $E = 2\mathbb{N}$, regularity and one-pebble recognition are established, while zero-pebble deterministic membership is undecided by the present results. Finally, if E is not ultimately periodic, $\mathcal{L}_E$ lies outside all standard finite-pebble tree-walking language classes because it is nonregular. The distinction between these three cases prevents the general regularity theorem from being mistaken for a DTWA membership decision.

A Finite-control realization details

This appendix records two physical implementation points used in the main text. It is part of the symbolic proof, not a claim that finite testing establishes a tree-walking lower bound.

A.1 Local virtual-head transitions for dilation

Fix $d \geq 2$. An original input node x is represented by (x, O). For every original internal node x, the $d-1$ wrapper nodes inserted on the edge from x to its right child are represented by (x, W_j) for $1 \leq j < d$ (numbered from top to bottom). The blank left leaf attached to wrapper j is represented by (x, D_j). This uses precisely $1 + 2(d-1)$ virtual-position tags. The virtual downward neighbors are

virtual position	left child	right child
(x, O), x internal	(x_L, O)	(x, W_1)
(x, W_j), $j < d-1$	(x, D_j)	(x, W_{j+1})
(x, W_{d-1})	(x, D_{d-1})	(x_R, O)

Their inverse parent operations are

virtual position	parent
(x, D_j)	(x, W_j)
(x, W_1)	(x, O)
(x, W_j), $j > 1$	(x, W_{j-1})
(x, O), $x \neq$ root, $x = x_{\text{par},L}$	$(x_{\text{par}}, \mathsf{O})$
(x, O), $x \neq$ root, $x = x_{\text{par},R}$	$(x_{\text{par}}, \mathsf{W}_{d-1})$

.

A virtual movement between original nodes induces the corresponding physical parent/child move; a movement inside a fixed original right-edge gadget only changes the finite tag. The virtual node label, root/left/right-child type, and leaf/internal type are all functions of the physical local type and tag. All virtual transitions can therefore be implemented by a bounded microprogram. For $d = 1$ only the tag O is needed, and the simulator is the original input walker. This supplies the promised deterministic implementation of Lemma 3.2 with no pebble or stack.

A.2 Anchored one-pebble probes and the monotone ascent

The one-pebble recognizer has a fixed sequence of phases: (i) check $\mathcal{W}_3$ by root-to-root DFS; (ii) root-to-root preorder with a saturated count of a leaves visited, interrupting at a candidate node for two bounded subtree probes; (iii) if q is found, abandon preorder and climb from q to r, probing left siblings only after an ascent along a right-child edge; and (iv) lift the pebble, if present, return to the root if required, and halt according to the final finite counter. Every subtree probe begins by placing the sole pebble at the candidate parent v and ends on the first return to v from the chosen child, when that pebble is lifted. Thus the ordinary preorder count and the probe count belong to disjoint phases and are not conflated.

The control can be organized into finitely many traversal phases: preorder entry, anchored left-child probe, anchored right-child probe, and upward ascent. A probe saves the saturated preorder count and the ascent summary in finite control; its own temporary marked-leaf count is reset when the pebble is placed and discarded after the return to that pebble. The ascent phase inspects the type of the child just left, which the standard tree-walking model exposes: leaving a left child continues upward without either testing the parent or incrementing the right-edge count; leaving a right child increments that count and starts a fresh anchored probe of the parent's left child. No probe remains active across successive ascent edges, and acceptance is tested only after an eligible right-child ascent reveals the leftmost marker. Thus there is no unbounded recursion or second pebble.

The test for q uses the preorder count 1 and child counts $(1, 1)$. For three marks, it can first succeed only at $q = \mathrm{lca}(a_2, a_3)$ in the right-pair case. During the climb, a positive left-sibling probe on a right-child ascent occurs first exactly at r; all lower such left siblings contain no marked leaf. Each qualifying ascent increments the finite E-counter once, and no backtracking probe changes that counter. Its final value is therefore the right-edge count on $r \rightsquigarrow q$. This justifies the one-pebble implementation independently of any bottom-up automaton construction.

A.3 Finite-instance consistency framework

The finite consistency analysis ranges over all ordered full binary tree shapes with $3 \le n \le 8$ leaves and all choices of three marked leaves. There are C_{n-1} such shapes with n leaves, where $C_j = \frac{1}{j+1}\binom{2j}{j}$ is the Catalan number. The exact number of marked inputs is therefore

$$\sum_{n=3}^{8} C_{n-1}\binom{n}{3} = 29{,}646.$$

For each input the statistic can be obtained directly from its two lowest common ancestors, or recursively from the child-count cases in Theorem 2.1. The search-and-ascent argument of Theorem 5.2 and the successful-turn bijection of Lemma 6.1 provide two further descriptions of the same integer. For $d = 1, 2, 3, 4$ the dilation identity of Lemma 3.1 gives $4 \cdot 29{,}646 = 118{,}584$ finite instances. These are transparent consistency bounds; the results for arbitrarily large trees rest on the symbolic proofs, not on the finite range.

Acknowledgments and proof provenance

The present arguments build on the published three-marker separation and the standard tree-walking closure principles identified in the references. The author's earlier monograph inspired the semantic-versus-physical viewpoint; it is cited as conceptual background and is not a premise of the lower-bound proofs. The finite-instance computations are independent consistency checks, not substitutes for symbolic arguments.

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